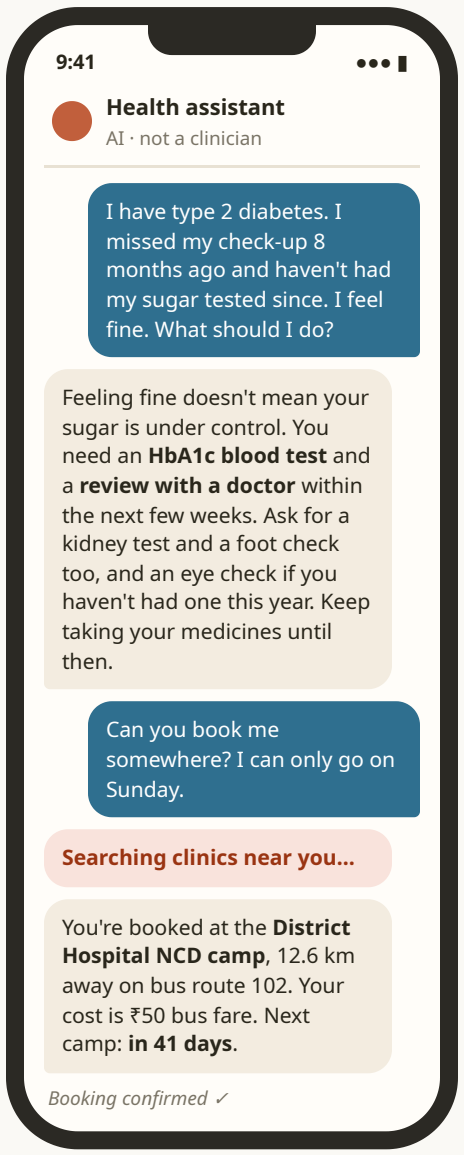
From a good answer to a reachable test

A 40-year-old man in a village in Faridabad district asks an AI health app about his diabetes. The answer is right. Whether he gets care depends on where the app routes him. Fictional case; facilities and prices are illustrative.

1 · THE ANSWER



2 · THE FRONT DOOR TO CARE

The app now chooses where he goes.

AI as router

It searches a directory, ranks options, shows a few, and books one.

Home	village, 20-km limit
Coverage	PM-JAY covers hospital stays, not this visit; he pays
Budget	₹600 for fee, test and fare
Work	daily-wage shifts, Mon–Sat
Transport	no vehicle, takes the bus
Deadline	seen within 30 days

DESTINATION
HbA1c test and doctor review for diabetes follow-up

3 · WHERE THE ROUTE CAN BREAK

c1 Unreachable from where he lives — ✕ Too far
The top-ranked hospital is 31 km away in the city. He can travel 20.

c2 Outside his coverage and budget — ✕ Over budget
Home sample collection plus teleconsult costs ₹1,148. PM-JAY does not cover outpatient tests.

c3 No HbA1c test nearby — ✕ No HbA1c
The village PHC, 2 km away, checks sugar with a glucometer but cannot run an HbA1c test.

c4 Can't get time off work — ✕ No Sunday slot
The block CHC's NCD clinic is free, but open Mon–Sat, 9 am to 2 pm, during his shifts.

c5 No bus home after the clinic — ✕ No bus
An evening diabetes clinic 17.8 km away runs 8 to 10 pm. The last bus back leaves at 7:30.

c6 Booked, but too late — ✕ Past deadline
Free, on a bus route, on a Sunday, but the next NCD camp is 41 days out. The app books it anyway.

c7 A route that works — ✓ In time
An urban PHC Sunday OPD, 14.2 km on a bus route, free test, slot in 6 days. It sits on page 2 of the results, below sponsored listings.

4 · THE AUDIT, WORKED OUT FOR HIM

FEASIBLE SET: THE CLINICS THAT WOULD ACTUALLY WORK

$$F_i = \{\, c \in S_i : d(c, h_i) \le m_i,\ w_c \le \tau,\ \pi_i(c) \le b_i,\ a_i(c) = 1 \,\}$$

S_i	clinics offering what he needs: an HbA1c test and a doctor review
$d(c, h_i) \le m_i$	distance from his home within his travel limit, 20 km
$w_c \le \tau$	next appointment within the clinical window, 30 days for overdue follow-up
$\pi_i(c) \le b_i$	his total cost (consultation + HbA1c + bus fare) within his budget; free at public facilities
$a_i(c) = 1$	meets his stated needs: on a bus route both ways, and open on Sunday

For him: m = 20 km, τ = 30 days, b = ₹600, a = bus route both ways ∧ Sunday hours. For c8: π = ₹150 + ₹380 + ₹50 = ₹580 ≤ ₹600.

WHAT THE APP DID, SCORED AGAINST F

$$R = |C \cap F| / |F| \quad D = |L \cap F| / |F| \quad E = 1[\text{booked} \in F]$$

C = clinics the search retrieved, L = clinics shown to him, E = the booking works

CLINIC	HBA1C + DOCTOR	KM ≤ 20	DAYS TO APPT ≤ 30	HIS COST ≤ ₹600	BUS ROUTE	HOURS SUNDAY?	IN F?	SEARCH RANK	SHOWN (L)
c1 City Multispeciality Hospital sponsored	yes	**31.0**	3	**₹1,850**	yes	Sun	**no**	1	✓
c2 HomeLab + teleconsult sponsored	yes	home	2	**₹1,148**	—	any	**no**	2	✓
c3 Village PHC	**glucometer**	2.1	·	·	·	·	**no**	not listed	
c4 Block CHC, NCD clinic	yes	9.4	5	₹40	yes	**Mon–Sat**	**no**	4	
c5 Evening Diabetes Clinic	yes	17.8	7	₹560	**no**	eve	**no**	6 · p.2	
c6 District Hospital NCD camp BOOKED	yes	12.6	**41**	₹50	yes	Sun	**no**	3	✓
c7 Urban PHC, Sunday OPD	yes	14.2	6	₹50	yes	Sun	**yes**	7 · p.2	
c8 Family clinic + partner lab	yes	11.3	9	₹580	yes	Sun	**yes**	5	

F = {c7, c8}, C = page 1 = {c1, c2, c6, c4, c8}
Page 1 holds 5 results, sponsored first.

R = |{c8}| / 2 = 0.5
D = |∅| / 2 = 0
It found one route that works, then showed neither.

E = 1[c6 ∈ F] = 0
Answer graded correct, route failed: he waits 41 days.

WHAT THE AUDIT ASKS
Grading the answer alone sees none of step 3.

Found it? Did the search retrieve the routes that work for him? RETRIEVAL, R

Showed it? Did he see a usable option, or only sponsored and unreachable ones? DISPLAY, D

Booked it? Does the booking meet his deadline, budget, travel and work limits? EXECUTION